2. Rooftop rainwater harvesting

It is a system of catching rainwater where it falls. In rooftop harvesting, the roof becomes the catchments, and the rainwater is collected from the roof of the house/ building. It can either be stored in a tank or diverted to artificial recharge system. This method is less expensive and very effective and if implemented properly helps in augmenting the groundwater level of the area.

Components of the Rooftop Rainwater Harvesting

The system mainly constitutes of following sub components:

Catchments

Transportation

First flush

Filter

Catchments

The surface that receives rainfall directly is the catchment of rainwater harvesting system. It may be terrace, courtyard, or paved or unpaved open ground.

Transportation

Rainwater from rooftop should be carried through down take water pipes or drains to storage/harvesting system. Water pipes should be UV resistant (ISI HDPE/ PVC pipes) of required capacity. Water from sloping roofs could be caught through gutters and down take pipe. At terraces, mouth of the each drain should have wire mesh to restrict floating material.

First Flush

First flush is a device used to flush off the water received in first shower. The first shower of rains needs to be flushed-off to avoid contaminating storable/rechargeable water by the probable contaminants of the atmosphere and the catchment roof. It will also help in cleaning of silt and other material deposited on roof during dry seasons Provisions of first rain separator should be made at outlet of each drainpipe.

Filter

Filters are used for treatment of water to effectively remove turbidity, colour and microorganisms. After first flushing of rainfall, water should pass through filters. A gravel, sand and 'netlon' mesh filter is designed and placed on top of the storage tank. This filter is very important in keeping the rainwater in the storage tank clean. It removes silt, dust, leaves and other organic matter from entering the storage tank. The filter media should be cleaned daily after every rainfall event.

INTERESTING FACT

Roof top rainwater is the most common practice in Shillong, Meghalaya. Eventhough Cherapunjee and Mawsynram situated at a distance of 55 km. from Shillong, receive heavy rainfall The state capital Shillong, faces acute shortage of water.Hence every household in the Shillong city has a roof top water harvesting structure.

Why water has to be purified?

If water is contaminated many diseases such as Hepatitis, Cholera, Diarrhoea, Typhoid fever, Dysentery, Amoebic dysentery results. Hence water has to be purified for consumption.

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